

EXÁMEN 2018

La estructura ABCD está sujeta a la acción de las siguientes cargas:

o Carga puntual P de 10 kN en C

o Incremento de T_a en la barra CD que es el exterior es 30° y el interior 10°

Desplazamiento horizontal de D

Reacciones

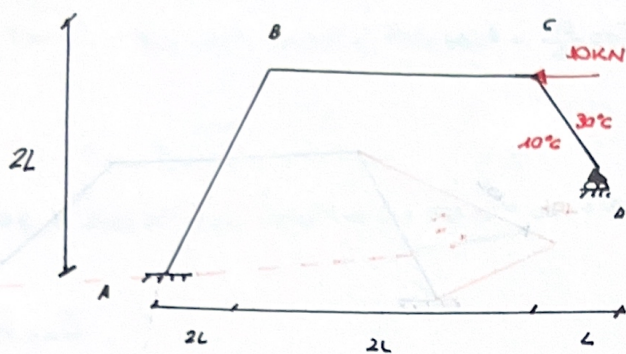
Distribución de flexiones.

En A y B .

$$EI = 500 \text{ kN} \cdot \text{m}^2$$

$$C = 1/10$$

$$\alpha = 10^{-5} / ^\circ\text{C}$$



$$\sin 45^\circ = \frac{2L}{d}$$

$$d = \frac{2L}{\sin 45^\circ} \rightarrow d = 2\sqrt{2}L$$

$$AB^2 = 2L^2 + 2L^2 \rightarrow AB^2 = 4L^2 \rightarrow AB = \sqrt{4L^2} \rightarrow AB = 2L$$

$$CD^2 = L^2 + L^2 \rightarrow CD = \sqrt{2L^2} \rightarrow CD = L\sqrt{2}$$

$$\sin 45^\circ = \frac{L}{d}$$

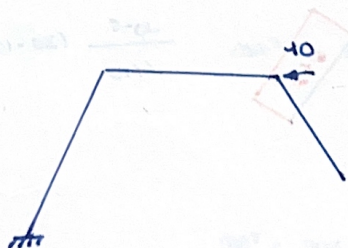
$$d = \sqrt{2}L$$

Hiperestática $GH = 1 \rightarrow [V_D = 0] \rightarrow \text{en compatibilidad.}$

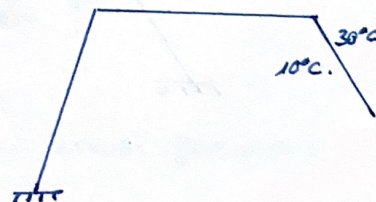
I)



II)

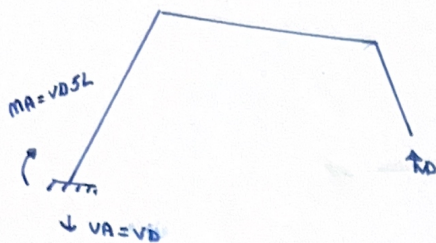


III)

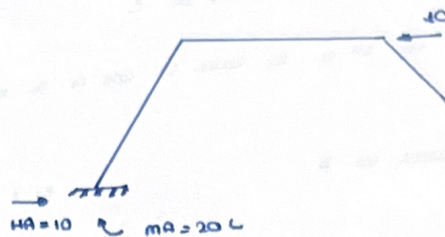


FLECTORES → reacciones en luces de la hipocética

I)

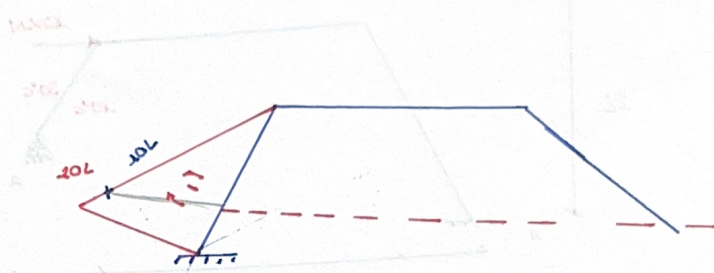
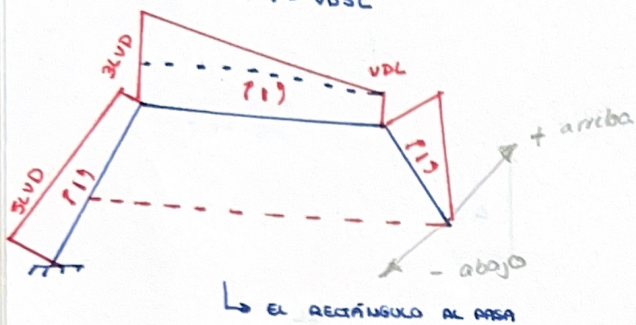


II)



$$MA - 5VDL = 0 \rightarrow MA = VDSL$$

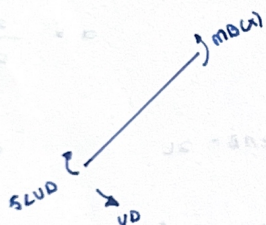
$$MA - 10 \cdot 2L = 0 \rightarrow MA = 20L$$



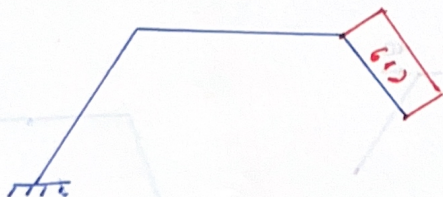
$$\bullet MB(L) - VDSL + 2VDL = 0$$

$$MB = VDSL - VD2L$$

$$[MB = 0VDL]$$



III) Estado térmico (fijación)



$$\frac{30-5}{2/10} (30-10) = \frac{1}{50000L} = M.$$

$$v_D = 0$$

xx NO OLVIDAR LA CONTRIBUCIÓN DEL INCREMENTO DE LONGITUD.

$$v_D = v_A + \frac{1}{EI} [A x d] + \text{Corr. térmica.}$$

$$0 = \frac{1}{EI} \left[3L \cdot v_D \cdot 2L \times \overbrace{(3L + L)}^{4L} + \frac{1}{2} v_{DL} \cdot 2L \times \overbrace{(2L + 2/3 \cdot 2L)}^{4L/3} + v_{DL} \cdot 2L \times 2L + \frac{1}{2} \cdot \frac{1}{L} \cdot 2 v_D \times \overbrace{(L + 2/3 \cdot 2L)}^{4L/3} + \frac{1}{2} \cdot \frac{1}{L} \cdot v_{DL} \cdot v_{DL} \times \frac{1}{3} L + \frac{1}{2} \cdot \frac{1}{L} \cdot v_{DL} \cdot 2L \times \overbrace{(3L + 2/3 \cdot 2L)}^{4L} \right] = \frac{1}{50000} \times \frac{52}{3} \times \frac{L}{2} = 10^{-5} \frac{(30+10)}{2}$$

→ LONGITUD EN LA DIRECCIÓN DEL

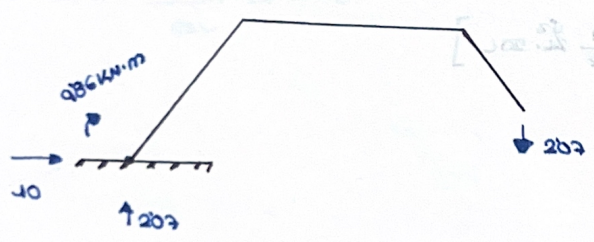
x DESPLAZAMIENTO → IMPORTANTE BARRAS INCLINADAS. → PROYECTAR EN ESA DIRECCIÓN

$$= \frac{1}{50000} \left[24 v_{DL}^3 + 867 v_{DL}^3 + 4 v_{DL}^3 + \frac{14}{3} v_{DL}^3 + \frac{52}{3} v_{DL}^3 + 9128 L^3 \right] = 14110^{-5} L - 28310^{-4} L$$

$$= 4'810^{-8} v_{DL}^3 + 1'73410^{-8} v_{DL}^3 + 810^{-4} v_{DL} + 9'3310^{-4} v_{DL} + 94310^{-5} v_{DL} + 0'01 v_{DL} = 2'9710^{-4} L$$

$$v_D = -2'07 \text{ kN}$$

Reacciones.



→ importante para localizar signo del momento

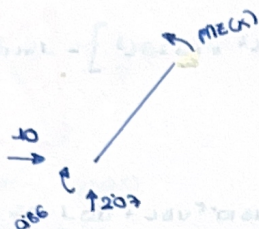
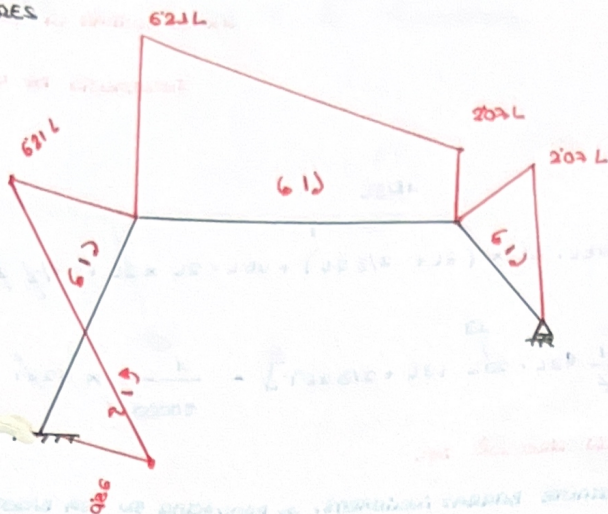
$$M_A^x \rightarrow M_A + 2'07 \cdot 5L = 0$$

para la M de referencia.

$$M_A = -1014$$

$$M_A^T = 20L = 1014 \rightarrow M_{TOTAL} = [986 \text{ kN.m}]$$

Diagrama de PECTORES



$$m(x) = 986 - 202 \cdot 2L + 10 \cdot 2L = 0$$

$$m(x) = 986 + 414L - 20L \rightarrow \text{más fácil sacar del}$$

diagrama de PECTORES de cada estado

Giro en B:

$$\vec{\phi}_B = \vec{\phi}_A + \frac{1}{EI} [A]$$

$$\vec{\phi}_B = \frac{1}{EI} \left[-3L \cdot 202L - \frac{1}{2} \cdot 2L \cdot 2L - \frac{1}{2} \cdot 2L \cdot 20L \right]$$

$$v_B = -202$$

$$\vec{\phi}_B = \frac{1}{5000} \left[-1926L^2 + 585L^2 - 2828L^2 \right]$$

$$[\vec{\phi}_B = -974 \cdot 10^{-4} \text{ rad}]$$

Desplazamiento horizontal en D. * EL ESE CORTA A LA ESTRUCTURA

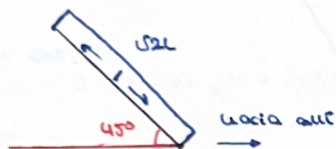
$$\vec{u}_D = \vec{u}_A + \frac{1}{EI} [A \times d] + \text{Contribución térmica.}$$

$$u_D = \frac{1}{EI} \left[\frac{1}{2} \cdot 52L \cdot VD \cdot L \times \frac{2}{3}L + VD \cdot L \cdot 2L \times L + \frac{1}{2} \cdot 2VDL \cdot 2L \times L + \frac{1}{2} \times \frac{252L}{2} \times 10L \times \frac{1}{3}L - \right.$$

$$\left. \frac{1}{2} \times \frac{252L}{2} \times 10L \times \frac{2}{3}L - \frac{252L}{2} \cdot 10L \times \frac{L}{2} + 3VD \cdot \frac{252L}{2} \times \frac{L}{2} + \frac{1}{2} \cdot \frac{252L}{2} \cdot VD \cdot L \times \frac{1}{3}L \right.$$

$$\left. - \frac{1}{2} \cdot \frac{252L}{2} \cdot VD \cdot L \times \frac{2}{3}L - 4VDL \cdot \frac{252L}{2} \times \frac{L}{2} \right] = \frac{1}{50000L} \times 52L \times \frac{1}{2} + 10 \cdot \frac{80+10}{2} \cdot \frac{1}{2} \cdot XL$$

$$u_D = -456 \cdot 10^{-2} L \text{ m}$$



$$\cos 45 = \frac{x}{52L} \rightarrow x = L.$$